SUMIT SINGH

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OBJECTIVE

AI/ML engineer experienced in language-model fine-tuning, retrieval, and production conversational systems. Seeking research and engineering opportunities to investigate model architectures, representation learning, and training methods.

EDUCATION

Master of Science in Applied Computing, *AI Concentration* Sep 2026 - Dec 2027
University of Toronto
Relevant Coursework: Neural Networks and Deep Learning; Imitation Learning for Robotics

Bachelor of Technology, Computer Science Engineering *(Specialization in AI & ML)* Jul 2018 - May 2022
SRM Institute of Science and Technology CGPA - 9.71/10.00

KEY SKILLS

Languages: Python, Go, C++, Java, SQL, JavaScript
ML/AI: Fine-tuning (LoRA/QLoRA), Quantization, NLP, RAG, Computer Vision, scikit-learn, XGBoost
Tools: PyTorch, Hugging Face Transformers, ChromaDB, NVIDIA NeMo Guardrails, Amazon Lex, Bedrock, AWS Lambda

PROFESSIONAL EXPERIENCE

Goldman Sachs — Associate (SDE-2) Jul 2022 - Jul 2026

• **Conversational AI-based IVR:** Developed and deployed an IVR using **Amazon Lex**, **AWS Lambda (Go)**, and Java microservices, replacing DTMF menus with conversational support across **60+ customer intents**; upgraded Lex from ASR 1.0 to **ASR 2.0** to improve transcription and intent classification for customer requests expressed in everyday language.
• **Intent Classification & Fine-tuning:** Analyzed production utterances to correct intent labels and build a fine-tuning dataset; fine-tuned **DistilBERT and Llama** and tested **LoRA/QLoRA** for Llama. Deployed quantized DistilBERT as an additional layer over **Amazon Lex**, using a smaller GPU to serve production traffic with lower inference latency and cost.
• **Customer Self-Service:** Improved the IVR's ability to resolve customer requests without agent transfers, contributing to containment growth from **10% to 75%** across **600K monthly calls** and **$5M+ in quarterly savings**.
• **Agent Assistance & Call Summarization:** Built a **speech-to-text RAG** assistant that retrieved relevant FAQs and documentation to suggest answers to agents; evaluated retrieval relevance and answer quality. Helped integrate **Amazon Bedrock** for handoff summaries. Together, the workflows saved **2 minutes per transferred call**, or an estimated **5K agent hours/month**.
• **Automated Bot Regression Testing:** Engineered a serverless workflow using **AWS Lambda and Step Functions** to process **3,000+ test utterances weekly**, generating reports on model and bot issues to identify regressions during bot updates.

Sabre — Associate Intern Jan 2022 - Jul 2022

• **Automated Python Data Scraping:** Developed Python-based web scraping scripts to collect airline data from public websites, and prepared the data in a structured format for downstream analysis with minimal manual intervention.
• **Airline Data Analytics:** Prepared and formatted airline data for an internal analytics and optimization tool, supporting analysis of flight scheduling, fleet allocation, and dynamic pricing for client airlines, with team and client reviews of the outputs.

PROJECTS

• **RF Capture Auto-labeler — Qoherent (Ongoing):** Co-developed a **U-Net + XCiT** pipeline to detect spectrogram regions and classify raw IQ into 53 known classes, fine-tuning RF-pretrained XCiT on synthetic data. Against the provided NB-det baseline on the same 500 recordings, detection F1 rose **21.39% to 54.44%**; end-to-end known-signal recall rose **6.45% to 22.64%**.

• **Imitation Learning for Robotic Manipulation — Course Project (Ongoing):** Investigating whether pairwise outcome-ranking supervision improves a world model's selection of action sequences proposed by a behavior-cloning policy trained on robot-lifting demonstrations. Planned MuJoCo evaluation includes cube positions held out from training to assess generalization.

• **PhysicsTeacher-CurrentElectricity-GPT** 🔗: Built a physics tutoring system for Grade 11 Current Electricity using **RAG and ChromaDB**, with textbook retrieval and **embedding-based conversation memory**. Used **OpenAI** generation and **NVIDIA NeMo Guardrails** for grounded, domain-constrained responses and context-aware follow-up handling.

• **ML-based Invoice Management Application** 🔗: Preprocessed **45K+** invoices, engineered **6 payment-trend features**, compared models, and tuned XGBoost via **5-fold cross-validation**, yielding **2.43-day MAE**, **0.61 R^2**.

• **Technical Writing:** Published 8 Python and machine learning tutorials on CodeSpeedy 🔗, with **11K+ views**.